STAFF REVIEW · H446 193 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. No. It is a copy.

2. A result a function passes to its caller.

3. 10.

4. 00010000, decimal 16.

I Do — source worked steps

1. Predict first — Write +9 in 8 bits and invert all the bits.

2. Model the result — Add 1 to the inverted bits: 11110111 represents −9.

We Do — guided CHECK

1. Predict first
+6 = 00000110. Invert to get 11111001.
Model the result
Add 1: 11111010 is the representation of −6.

You Do — source mark schemes

1. +5=00000101; inverted=11111010; add 1=11111011.

2. 00011010, which is 26 (52 / 2).

3. 11111010, representing −6. The sign bit 1 is copied into the new leftmost position.

Exit Ticket — source answer

1. False. An arithmetic right shift replicates the sign bit (1 for negative values).